Hyperspectral Object Tracking With Dual-Stream Prompt

Rui Yao, Lu Zhang, Yong Zhou, Hancheng Zhu, Jiaqi Zhao, *Member, IEEE*, and Zhiwen Shao

***Abstract*— Hyperspectral images, rich in spectral details, offeradvantages for object tracking across diverse scenarios. Current hyperspectral tracking often fine-tunes parameters using pretrained RGB trackers, but this manner is suboptimal due to redundancy in spectral bands and limited training data. Existing hyperspectral trackers also underuse temporal information. To address these issues, we propose a unified spectral–spatiotemporal multimodal dual-stream prompt hyperspectral object tracking, named HDSP. We design a density clustering-based band selection module (BSM) to preserve spectral prompt information efficiently. Using the generated bands and temporal data as multimodal prompts, a dual-stream visual prompter is proposed. Designed multimodal dual-stream visual prompter (MDVP) transforms the multimodal input into a single modality, enhancing the foundational modality's representation capabilities for hyperspectral tracking. Experiments on hyperspectral videos (HSVs) tracking datasets demonstrate that the proposed tracker achieves state-of-the-art performance. The source code is available at https://github.com/rayyao/HDSP.**

***Index Terms*— Band selection, dual-stream prompt, hyperspectral object tracking, hyperspectral video (HSV).**

I. Introduction

VISUAL object tracking is a crucial task in computer vision, which has undergone significant development and widespread application in various fields over the past decade. It has been successfully applied in numerous applications such as video surveillance [1], artificial intelligence [2], face tracking [3], and autonomous driving [4]. Recently, with the continuous development of spectral imaging technology, hyperspectral videos (HSVs) can be obtained by hyperspectral snapshot sensors [5], which makes the algorithm of object tracking using HSVs gradually popular [6]. Compared with RGB images, multiband hyperspectral images can provide detailed scene content composition information, which enhances the ability to distinguish objects from complex backgrounds, can effectively solve challenges such as object deformation and occlusion, and are conducive to object tracking.

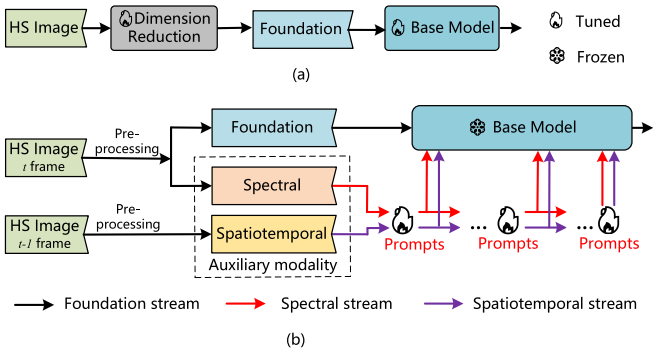


Fig. 1. Illustration of different hyperspectral tracking paradigms. (a) Fine–tuning parameters of the RGB tracker using hyperspectral images followed by dimension reduction. (b) Proposed multimodal dual-stream prompt hyperspectral object tracking (HDSP). HDSP effectively incorporates spectral and spatiotemporal prompts, leading to more accurate localization of tracked object.

The early material-based hyperspectral tracker [5] used manual feature extractors to capture material information from HSVs. It used background-aware correlation filters (BACFs) for object tracking. Recent hyperspectral tracking research has adopted deep learning features. Due to the limited training samples in hyperspectral object tracking, deep-learning-based hyperspectral trackers rely on the existing pretrained RGB trackers. However, these trackers cannot leverage hyperspectral semantic features, resulting in poor generalization and reduced prediction accuracy, which ultimately limits the precision of bounding boxes generated by hyperspectral trackers.

In addition, since hyperspectral images have multispectral characteristics, there is redundancy between spectral, and the channels of hyperspectral images do not match RGB images, it is not feasible to directly use pretrained RGB trackers for tracking. Common solution is to convert hyperspectral images into three-channel images through different dimensionality reduction methods. After this conversion, comprehensive fine-tuning is done on RGB-based parameters [see Fig. 1(a)]. The existing hyperspectral dimensionality reduction methods [7], [8], [9], [10], [11], [12] are mainly divided into two categories. The first category of methods reduces hyperspectral images to single three-band image [7], [8], [9]. For example, BAHT [7] and SiamHT [8] use background-aware band selection module (BSM) and heterogeneous encoder–decoder module, respectively, to map the hyperspectral image into a single three-channel image. SiamHYPER [9] selects three

Received 22 April 2024; revised 17 July 2024, 17 August 2024, 29 September 2024, and 9 November 2024; accepted 10 December 2024. Date of publication 13 December 2024; date of current version 20 December 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62172417, Grant 62272461, Grant 62101555, Grant 62106268, and Grant 62472424; in part by Xuzhou Key Research and Development Program under Grant KC22287; and in part by the "Double First-Class" Project of China University of Mining and Technology for Independent Innovation and Social Service under Grant 2022ZZCX06. *(Corresponding author: Rui Yao.)*

The authors are with the School of Computer Science and Technology and the Mine Digitization Engineering Research Center of the Ministry of Education, China University of Mining and Technology, Xuzhou 221116, China (e-mail: ruiyao@cumt.edu.cn; ts22170030a31@cumt.edu.cn; yzhou@cumt.edu.cn; zhuhancheng@cumt.edu.cn; jiaqizhao@cumt.edu.cn; zhiwenshao@_cumt.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3516833

1558-0644 © 2024 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:06:08 UTC from IEEE Xplore. Restrictions apply.

bands from hyperspectral data and inputs them into a backbone network pretrained on color videos. The second category of methods reorganizes bands into multiple sets of three-band images for tracking, as in BAE-Net [10], CBFF-Net [11], and SiamBAG [12]. These methods regroup bands based on various criteria and feed them into pretrained tracking networks to generate score maps, but this can lead to information redundancy and higher computational complexity.

Current solutions for hyperspectral tracking still have significant limitations.

1) Hyperspectral images' band redundancy. Hyperspectral images contain more bands than RGB images, leading to interference and inefficiency during tracking due to duplicated information.
2) Limited hyperspectral training data. Scarcity of real-time, high-quality hyperspectral data from the existing sensors hinders effective transfer training for RGB networks, compromising the robustness of hyperspectral object tracking.
3) Temporal information underutilization. Conventional hyperspectral trackers often overlook crucial temporal changes in HSVs, impacting their ability to handle complex scenarios such as rapid object changes and occlusion.

Hence, we propose HDSP, a unified spectral–spatiotemporal multimodal dual-stream prompt hyperspectral object tracking method. First, we designed a band selection method based on density clustering to tackle the redundancy of hyperspectral band information, effectively eliminating interband redundancy while retaining rich spectral information. Recently, models using prompt learning have emerged in the multimodal domain, such as visual prompt-based multimodal tracking [13], [14]. To address the limited hyperspectral training data issue, we adopted the concept of prompt learning, using a pretrained RGB tracker as the base model for hyperspectral object tracking. HDSP freezes the entire base model and only learns modality-specific visual prompts, avoiding the need for fully fine-tuning the RGB tracker and auxiliary modality branch, thus maximizing the inheritance of RGB tracker parameters. However, compared with RGB videos, HSVs with NN bands need to undergo dimensionally reduced to be applicable to the RGB tracker. Therefore, we used a density clustering-based band selection method to select three representative bands, and the pseudocolor image composed of the selected three bands is input into the tracker as spectral prompt information. Furthermore, to address inadequate utilization of temporal information, in addition to spectral prompts, we used interframe data as spatiotemporal prompts, inputting the false-color image as base information into the backbone network. The multimodal dual-stream visual prompter (MDVP) then combines the base information with the spectral and spatiotemporal prompts [see Fig. 1(b)], enhancing the expressiveness of the base information and improving tracking performance.

The contributions of our article are as follows.

1) We propose an end-to-enddual-stream prompt-based hyperspectral object tracking framework, incorporating auxiliary modal inputs for spectral and spatiotemporal information. Spectral prompts aid in distinguishing object from complex backgrounds, while spatiotemporal information enables rapid and dynamic adaptation to changing object appearances. Experiments on the HSV dataset demonstrate our tracker's excellent performance.
2) We devise a band selection method based on density clustering, wherein redundant bands are eliminated, retaining only those with minimal correlation and maximal information content as spectral prompt information.
3) We propose an efficient MDVP that uses attention mechanisms to adaptively blend spectral and spatiotemporal information through multiple cross correlation prompt layers (CPLs). These improved features are added to the base modality for efficient adaptation to downstream tasks.

II. Related Work

A. Visual Object Tracking

Object tracking methods are primarily categorized into correlation filtering and deep learning methods. The discriminative correlation filter (DCF) [15] addresses tracking by solving ridge regression in the Fourier domain, while the kernelized correlation filter (KCF) [16] maps image features into high-dimensional space using a kernel function for tracking. Recently, Siamese-based trackers have gained attention for their precision and efficiency. SiamGAT [17] applies graph attention to manage complex object relationships, and SiamBAN [18] introduces the boundary-aware network (BAN) for multiscale object information. Simultaneously, transformers [19] have brought significant performance improvements to tracking tasks. DETR [20] pioneers the use of the transformer for object detection. Inspired by DETR, STRAK [21] introduces the encoder–decoder transformer to enhance the performance of convolutional models. TransT [22] and SparseTT [23] enhance the interaction between template and search region features using transformers. OSTrack [24] introduces a single-stream framework unifying feature learning and relationship modeling. ARTrack [25] treats tracking as a coordinate sequence interpretation task, enabling cross-frame tracking through a simple end-to-end model. Seqtrack [26] models tracking as object coordinate sequence generation, achieving strong performance on multiple datasets. In addition, to overcome the speed limitations of visual trackers, FEAR family [27] proposes a dual-template representation and pixelwise fusion. HCAT [28] introduces an efficient tracking method using a hierarchical cross-attention transformer. E.T. Track [29] introduces a transformer module with a single instance-level attention layer for real-time tracking. HiT [30] includes a bridge module to enhance shallow features with deep information and uses dual-image position encoding. These methods can run at high speed on different devices while maintaining high performance. Although color trackers have made significant progress, their inherent limitations in capturing physical details can lead to drift in complex scenes due to color similarity with the background or lighting changes. In contrast, the object tracking based on HSVs provides more

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:06:08 UTC from IEEE Xplore. Restrictions apply.

adequate spatial and spectral information, which is conducive to improving the tracking accuracy.

B. Hyperspectral Object Tracking

Early hyperspectral object tracking methods used manual feature engineering. For example, CNHT [31] proposes fixed convolution filters by extracting normalized 3-D cubes from object regions. MFI [32] uses HOG and a pretrained VGGNet network to generate feature maps, detecting object in HSV with a kernel correlation filter framework. Zhang et al. [33] design a band selection strategy for HSVs and inputs the selected bands into the BACF [34] tracking framework, which successfully reduces the massive input of HSVs. However, manually crafted features have limitations, leading to tracking drift. Current hyperspectral trackers leverage pretrained backbone networks on color images or videos. For instance, SiamHYPER [9] uses pretrained Siamese tracking for tracking after dimensionality reduction. DeepHKCF [35] reduces hyperspectral data to three channels, performing deep feature extraction using VGGNet pretrained on RGB data. BS-SiamRPN [36] uses optimization algorithms to determine informative bands and applies transfer learning on a pretrained Siamese network with limited hyperspectral information. CBFF-Net [11] divides images into five groups for feature fusion and tracks them through pretrained TransT [22]. TFTN [37] and TDFF [38] use transformer to fuse hyperspectral information and RGB information to implement hyperspectral tracking methods. SEE-Net [39] establishes a spectral self-expression model to learn band correlations and generated weighted pseudocolor images for robust object tracking. Su et al. [40] propose to sort the hyperspectral bands by their Gaussian information entropy (GIE) values and select the three bands with the highest GIE values. The selected bands are then fed into SiamBAN for unsupervised tracking. SiamCAT [41] introduces a channel adaptive dual Siamese network for hyperspectral object tracking with varied channels. Trans-DAT [42] proposes a domain adaptive-aware hyperspectral object tracking framework. Despite progress, challenges such as band redundancy and poor transferability persist, and the temporal–spatial relationships have been overlooked in previous object trackers. These relationships are crucial for handling rapid motion and deformation of object.

C. Visual Prompt Learning

Fine-tuning has traditionally been the go-to method for leveraging large pretrained models in downstream tasks, but it is considered inefficient. Prompt learning offers an alternative by freezing backbone parameters and introducing a small number of learnable parameters. It has demonstrated performance comparable to full fine-tuning, as shown by VPT [43], gaining recognition in computer vision. Prompt learning's effectiveness is evident in tasks such as AIM's [44] remarkable video action recognition with minimal trainable parameters. ProTrack [14] combines multimodal prompts with STARK [21] for strong performance. ViPT [13] significantly improves tracking robustness by integrating limited multimodal data into extensively pretrained baseline models. MVTTG [45] uses image descriptions as valuable supplementary information and rich feature representation to improve tracking accuracy and robustness, solving the limitations of the existing multimodal tracking methods that only focus on visual data. Unlike previous single-stream prompt methods, we introduce prompts from dual-stream modalities: spectral and spatiotemporal information. HDSP effectively integrates both, resulting in more accurate object localization.

III. Method

A. Overview

As mentioned above, directly applying models pretrained on RGB data to hyperspectral object tracking is challenging. Therefore, we use a prompt learning mechanism to achieve effective transfer of hyperspectral object tracking. The network architecture, shown in Fig. 2, uses ViT [46] and OSTrack [24] as the base models.

1) Input Stream: The input to the HDSP network consists of adjacent frames of hyperspectral images for time t and $t-1$, denoted as $\mathbf{H}_t \in \mathbb{R}^{N\times H\times W}$ and $\mathbf{H}_{t-1}$. Here, N is the number of bands. H and W represent the height and width of the input image, respectively. Specifically, for the tth frame image $\mathbf{H}_t$, we obtain the false-color image $\mathbf{F}_t$ corresponding to the tth frame using (20) of CIE CMFs (see Appendix). Next, three representative bands are obtained through the BSM for the tth frame, forming the pseudocolor image $\mathbf{P}_t \in \mathbb{R}^{3\times H_c\times W_c}$. Simultaneously, for the $t-1$ frame, $\mathbf{H}_{t-1}$ inputs CIE CMFs to obtain the false-color image $\mathbf{F}_{t-1} \in \mathbb{R}^{3\times H_c\times W_c}$. In this manner, these three images are provided to the patch embedding layer, where each input stream is mapped and flattened into the latent space to generate corresponding tokens. Here, we denote the initial tokens generated from $\mathbf{F}_t$, $\mathbf{P}_t$, and $\mathbf{F}_{t-1}$ as $\widetilde{\mathbf{F}}_t^0$, $\widetilde{\mathbf{P}}_t^0$, and $\widetilde{\mathbf{F}}_{t-1}^0$.

2) Multimodal Dual-Stream Visual Prompter: Next, we input the aforementioned initial tokens into MDVP, which consists of an initial CPL (ICPL) and multiple subsequent CPLs. On one hand, ICPL consists of a motion feature fusion module (MFFM) and two prompt generation modules (PGMs). MFFM fuses the base information of the current frame with the previous frame to obtain spatiotemporal prompt $\widetilde{\mathbf{F}}_{t-1}^0$. On the other hand, CPL module consists of two PGMs. Thus, ICPL and CPL can use two PGMs to combine the spectral prompt $\widetilde{\mathbf{P}}_t^{l-1}$, the spatiotemporal prompt $\widetilde{\mathbf{F}}_{t,t-1}^{l-1}$, and the foundation information $\mathbf{F}_t^{l-1}$ to generate spectral and spatiotemporal prompt for layer l.

3) Base Model: We elementwise add the spectral prompt $\widetilde{\mathbf{P}}_t^l$, the spatiotemporal prompt $\widetilde{\mathbf{F}}_{t,t-1}^l$, and the foundation information $\widetilde{\mathbf{F}}_t^{l-1}$ of the previous layer into the lth layer encoder. The result of the final encoder layer undergoes channel dimension reduction through a linear layer. Subsequently, it is fed into the prediction head for classification and object bounding box regression. For detailed information on the head's localization, refer to OSTrack [24]. Note that in HDSP, all the network parameters related to the RGB modality are frozen, including patch embedding, feature extraction and interaction, and prediction head.

B. BSM Based on Density Clustering

Hyperspectral sensors capture reflection values across various wavelength bands to cover the entire spectral range.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:06:08 UTC from IEEE Xplore. Restrictions apply.

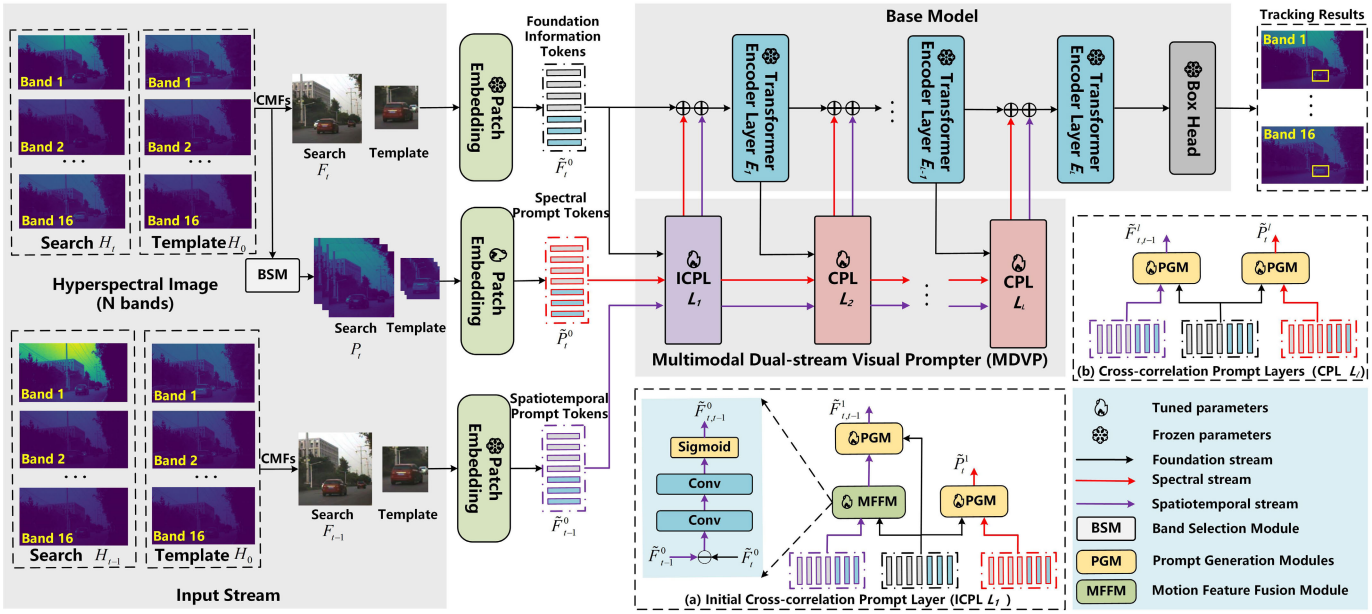


Fig. 2. Architecture of HDSP. HDSP is mainly composed of input stream, MDVP, and base model. The role of input stream is to preprocess the input hyperspectral image, involving CIE CMFs and BSM, to generate $\mathbf{F}_t$, $\mathbf{P}_t$, and $\mathbf{F}_{t-1}$. Subsequently, they are fed into patch embedding to obtain corresponding foundation and prompt tokens. These tokens enter the MDVP, including the ICPL and subsequent CPL, to generate effective visual prompts added to the foundation modality. Finally, feature extraction and interaction are achieved using base model (a multilayer stacked ViT backbone), ultimately realizing the tracking process. (a) ICPL L_1. (b) CPL L_1.

Inappropriately increasing spectral bands not only fails to add new information but also adds complexity to tracking tasks. Therefore, we designed a BSM based on density clustering. The BSM uses the image entropy of each band for density clustering to obtain local density values. Subsequently, it sequentially calculates the minimum distance between the image entropy of each band and the image entropy of bands with higher local density to determine the correlation between bands. BSM selects the three bands with the lowest correlation, which can effectively removes redundancy in bands while preserving essential spectral information. The algorithm is shown in Algorithm 1.

1) Image Entropy: We calculate the image entropy to evaluate the information content of the ith band $\mathbf{H}_t^i$ in the hyperspectral image, where $i \in \{1, \ldots, N\}$. Specifically, given the image band $\mathbf{H}_t^i$, its image entropy can be defined as

$$E\left(\mathbf{H}_t^i\right) = -\int p\left(\mathbf{H}_t^i\right) \log p\left(\mathbf{H}_t^i\right) d\mathbf{H}_t^i. \tag{1}$$

Here, $p(\cdot)$ represents the probability density function of the band, which can be obtained by normalizing its pixel intensity histogram. Then, we map the band sequence i to the horizontal axis and $E(\mathbf{H}_t^i)$ to the vertical axis in a 2-D space, as shown in Fig. 3(a).

2) Local Density and Minimum Distance: Next, we perform density clustering [47] of the bands using the image entropy $E(\mathbf{H}_t^i)$ of each band. In addition, we sequentially choose each band as the clustering center and calculate the local density value $\rho(E(\mathbf{H}_t^i))$ and the minimum distance $\delta(E(\mathbf{H}_t^i))$ between each center. Specifically, the local density value depends on the direct ℓ_2 distance between each image entropy. Assuming it satisfies the triangle inequality, the local density is

Algorithm 1 Band Selection Algorithm Based on Density Clustering

Input: Hyperspectral image $\mathbf{H}_t^i$ with N bands, where $i \in \{1, \ldots, N\}$

Output: Pseudocolor image $\mathbf{P}_t$ with 3 bands

```
for i = 1 to N do
    Calculate image entropy E(H_t^i) using (1).
end for
Map E(H_t^i) to a 2-D coordinate space.
for i = 1 to N do
    Calculate local density ρ(E(H_t^i)) using (2).
end for
for i = 1 to N do
    Calculate minimum distance δ(E(H_t^i)) using (3).
end for
Map ρ and δ into a 2-D coordinate space.
Select the first 3 clustering centers with the highest δ values.
Combine these 3 bands to form P_t.
return P_t
```

defined as

$$\rho\left(E\left(\mathbf{H}_t^i\right)\right) = \sum_{j=1}^{N} \varphi\left(\ell_2\left(E\left(\mathbf{H}_t^i\right), E\left(\mathbf{H}_t^j\right)\right) - d_r\right) \tag{2}$$

where $\varphi(x)$ equals 1 if $x < 0$, and 0 otherwise. d_r is the clustering radius, and it is determined by one-third of the difference between the maximum and minimum values of the image entropy of the N bands in H_t, where the divisor is the same as the number of channels in RGB. If the distance

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:06:08 UTC from IEEE Xplore. Restrictions apply.

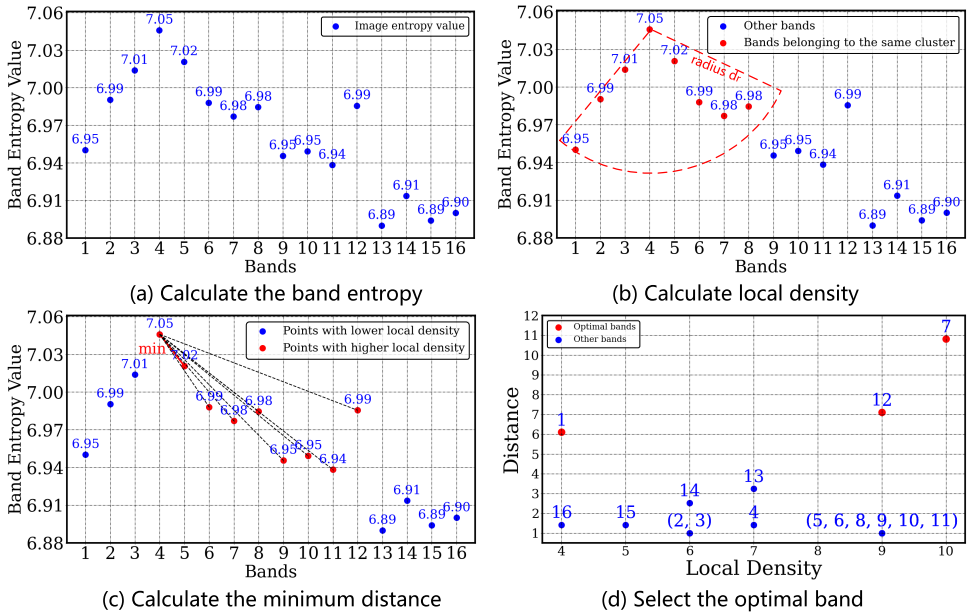


Fig. 3. Describing the BSM process using the first frame of the *ball* HSV (with N as 16). (a) Entropy graphs for various bands with blue numbers indicating the entropy. (b) Red circle with the entropy of the fourth band as the center and d_r as the radius. Red points inside the circle belong to the same category as the fourth. (c) Distance graph of the image entropy between the fourth band and the band with higher local density (red is the minimum distance). (d) Clustering decision graph with the optimal band points (red).

between the information entropy of two bands is negative, they belong to the same category, as shown in Fig. 3(b).

We calculate the minimum distance of information between each pair of bands based on their respective local density values, defined as follows:

$$\delta\left(E\left(\mathbf{H}_t^i\right)\right) = \min_{E(\mathbf{H}_t^j):\rho(E(\mathbf{H}_t^j))>\rho\left(E\left(\mathbf{H}_t^i\right)\right)} \ell_2\left(E\left(\mathbf{H}_t^i\right), E(\mathbf{H}_t^j)\right). \quad (3)$$

Fig. 3(c) illustrates the minimum distance.

3) Optimal Bands: We use (2) and (3) to obtain ρ and δ for each band, and then map them into a 2-d coordinate system. To maintain consistency with the three channels of RGB and basic information, we select the first three clustering centers with the highest δ values, as shown in Fig. 3(d). We combine these three bands to form a three-channel pseudo-color image $\mathbf{P}_t$, which serves as the input for spectral prompt information into the network.

C. Multimodal Dual-Stream Visual Prompter

The proposed MDVP inserts prompt streams into multiple stages of the base network, which consists of an ICPL and multiple CPLs, as shown in Fig. 2(a) and (b). Specifically, ICPL is defined as

$$\widetilde{\mathbf{P}}_t^1, \widetilde{\mathbf{F}}_{t,t-1}^1 = \texttt{ICPL}\left(\widetilde{\mathbf{P}}_t^0, \widetilde{\mathbf{F}}_{t-1}^0, \widetilde{\mathbf{F}}_t^0\right) \quad (4)$$

where the outputs $\widetilde{\mathbf{P}}_t^1$ (spectral information) and $\widetilde{\mathbf{F}}_{t,t-1}^1$ (spatiotemporal information) of ICPL are the dual-stream visual prompt of the next layer of the base network. Similarly, the definition of the lth layer of CPL is

$$\widetilde{\mathbf{P}}_t^l, \widetilde{\mathbf{F}}_{t,t-1}^l = \texttt{CPL}_l\left(\widetilde{\mathbf{P}}_t^{l-1}, \widetilde{\mathbf{F}}_{t,t-1}^{l-1}, \widetilde{\mathbf{F}}_t^{l-1}\right) \quad (5)$$

where $l \in \{2, \ldots, L\}$, and L is the number of layers in MDVP. We elementwise add the spectral–spatiotemporal dual-stream prompt to the foundation stream token $\mathbf{F}_t^{l-1}$ and input it into the lth layer encoder of the base network

$$\widetilde{\mathbf{F}}_t^l = \texttt{Encoder}_l\left(\widetilde{\mathbf{P}}_t^l + \widetilde{\mathbf{F}}_{t,t-1}^l + \widetilde{\mathbf{F}}_t^{l-1}\right). \quad (6)$$

Thus, the encoder token of the basic network of each layer can be generated.

1) ICPL and CPL: To capture the spatiotemporal features of hyperspectral objects, we introduce an MFFM within ICPL. Initially, we compute the absolute difference between the input $\widetilde{\mathbf{F}}_t^0$ and $\widetilde{\mathbf{F}}_{t-1}^0$ to obtain the frame difference $\widehat{\mathbf{F}}_t$

$$\widehat{\mathbf{F}}_t = \left|\widetilde{\mathbf{F}}_t^0 - \widetilde{\mathbf{F}}_{t-1}^0\right|. \quad (7)$$

Then we use two convolutional layers plus a Sigmoid activation function to obtain the initial spatiotemporal prompt information $\widetilde{\mathbf{F}}_{t,t-1}^0$ [shown in Fig. 2(a)]

$$\widetilde{\mathbf{F}}_{t,t-1}^0 = \texttt{Sigmoid}\left(\texttt{Conv}(\texttt{Conv}(\widehat{\mathbf{F}}_t))\right). \quad (8)$$

We input $\widetilde{\mathbf{F}}_{t,t-1}^0$ and $\widetilde{\mathbf{F}}_t^0$ into PGM to obtain the first layer of spatiotemporal prompt information. Simultaneously, $\widetilde{\mathbf{P}}_t^0$ and $\widetilde{\mathbf{F}}_t^0$ are input into PGM to obtain the first layer of spectral hint information $\widetilde{\mathbf{P}}_t^1$. The structure of the CPL layer is similar to ICPL, but CPL consists of only two PGMs.

2) PGM: To generate effective visual prompts for tracking, we design PGM. As shown in Fig. 4, PGM consists of the following three parts: channel attention module (CAM), spatial attention module (SAM), and cross-fusion module (CFM). CAM highlights the importance of different channels in object positioning by adjusting the weight of each feature channel. By calculating the attention weight of features in the spatial dimension, SAM identifies key areas in different modalities and enhances the object positioning ability of the model. CFM is used to achieve deeper information fusion.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:06:08 UTC from IEEE Xplore. Restrictions apply.

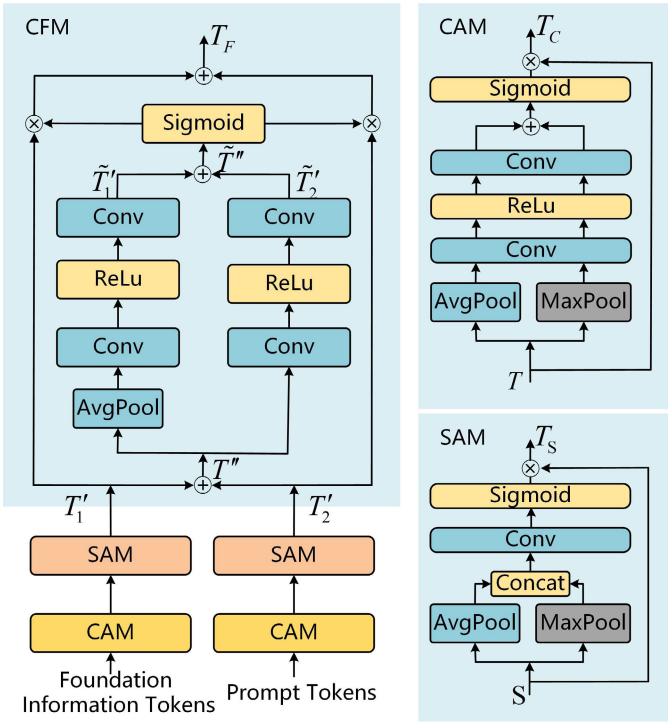


Fig. 4. Detailed design of the proposed PGM. The foundation information tokens and the prompt tokens (spectral prompt tokens or spatiotemporal prompt tokens) are input into the CAM and SAM for tokens' enhancement, and then cross-fused in the CFM to generate the final prompt.

For the CAM, assuming the input tokens are represented by $\mathbf{T} \in \mathbb{R}^{D\times H_k\times W_k}$, we perform average pooling and max pooling along the $H_k \times W_k$ dimension

$$\mathbf{Z}^c_{\text{avg}} = \texttt{AvgPool}(\mathbf{T}), \quad \mathbf{Z}^c_{\text{max}} = \texttt{MaxPool}(\mathbf{T}). \tag{9}$$

Next, we input $\mathbf{Z}^c_{\text{avg}}$ and $\mathbf{Z}^c_{\text{max}}$ into the 1×1 convolution layer for channel dimensionality reduction, then pass the resulting features through the ReLU layer for nonlinear enhancement, and then use another 1×1 convolutional layer and add the Sigmoid activation function to project back to the original dimension. Thus, we can generate a weight map $\mathbf{Z}_{\text{channel}}$ with the same dimension as the token $\mathbf{T}$

$$\begin{aligned}\mathbf{Z}_{\text{channel}} = \texttt{Sigmoid}\big[&\texttt{Conv}\big(\texttt{ReLU}\big(\texttt{Conv}(\mathbf{Z}^c_{\text{avg}})\big)\big)\\ &+ \texttt{Conv}\big(\texttt{ReLU}\big(\texttt{Conv}\big(\mathbf{Z}^c_{\text{max}}\big)\big)\big)\big].\end{aligned} \tag{10}$$

We multiply the obtained $\mathbf{Z}_{\text{channel}}$ by the original token $\mathbf{T}$ to obtain the channel attention token $\mathbf{T}^c$

$$\mathbf{T}^c = \mathbf{T} * \mathbf{Z}_{\text{channel}}. \tag{11}$$

For the SAM, assuming the input tokens are represented by $\mathbf{S} \in \mathbb{R}^{D\times H_k\times W_k}$, we perform average pooling and max pooling along the D dimension to generate $\mathbf{Z}^s_{\text{avg}}$ and $\mathbf{Z}^s_{\text{max}}$. Then, we concatenate them along the D dimension and input them into a $(H_k - 1) \times (W_k - 1)$ convolutional layer and add the Sigmoid activation function to obtain the weight map

$$\mathbf{Z}_{\text{spatial}} = \texttt{Sigmoid}\big[\texttt{Conv}\big(\texttt{Concat}\big(\mathbf{Z}^s_{\text{avg}}, \mathbf{Z}^s_{\text{avg}}\big)\big)\big]. \tag{12}$$

We multiply the obtained $\mathbf{Z}_{\text{spatial}}$ by the original token $\mathbf{T}$ to obtain the spatial attention token

$$\mathbf{T}^s = \mathbf{S} * \mathbf{Z}_{\text{spatial}}. \tag{13}$$

We sequentially use (11) and (13) to obtain the attention-enhanced token $\mathbf{T}'$, as shown in Fig. 4.

For the CFM, we cross-fuse two tokens $\mathbf{T}'_1$ and $\mathbf{T}'_2$ to obtain the prompt token. We first add $\mathbf{T}'_1$ and $\mathbf{T}'_2$ to obtain $\mathbf{T}''$

$$\mathbf{T}'' = \mathbf{T}'_1 + \mathbf{T}'_2. \tag{14}$$

Then, we input $\mathbf{T}''$ into two different network branches on the left and right, respectively, for operation. For the left branch, $\mathbf{T}''$ undergoes average pooling, followed by a 1×1 convolutional layer, then ReLU activation, and then another 1×1 convolutional layer to obtain $\widetilde{\mathbf{T}}'_1$

$$\widetilde{\mathbf{T}}'_1 = \texttt{Conv}\big[\texttt{ReLU}\big(\texttt{Conv}(\texttt{AvgPool}(\mathbf{T}''))\big)\big]. \tag{15}$$

For the right branch, $\mathbf{T}''$ directly undergoes a 1×1 convolutional layer, then ReLU activation, and then another 1×1 convolutional layer to obtain $\widetilde{\mathbf{T}}'_2$

$$\widetilde{\mathbf{T}}'_2 = \texttt{Conv}\big[\texttt{ReLU}\big(\texttt{Conv}(\mathbf{T}'')\big)\big]. \tag{16}$$

Then, we add $\widetilde{\mathbf{T}}'_1$ and $\widetilde{\mathbf{T}}'_2$ to obtain $\widetilde{\mathbf{T}}''$

$$\widetilde{\mathbf{T}}'' = \widetilde{\mathbf{T}}'_1 + \widetilde{\mathbf{T}}'_2. \tag{17}$$

Finally, we apply the Sigmoid function to calculate the weights for $\mathbf{T}'_1$ and $\mathbf{T}'_2$, balance their importance by setting the weight sum to 1, and then obtain $\mathbf{T}^F$ through weighted summation

$$\mathbf{T}^F = \mathbf{T}'_1 * \texttt{Sigmoid}(\widetilde{\mathbf{T}}'') + \mathbf{T}'_2 * \big[1 - \texttt{Sigmoid}(\widetilde{\mathbf{T}}'')\big]. \tag{18}$$

D. Training of HDSP

HDSP is initialized with parameters from the base model OStrack [24]. For efficiency, we directly add the prompt information to the original token. This approach does not increase the number of tokens while achieving better performance. During the training process, data propagate throughout the entire model. However, for dual-stream visual prompt learning, we only need to update the gradient values of a few parameters for fast learning. Thus, the proposed model can achieve convergence in a short time, which effectively uses the rich knowledge in the pretrained basic model to narrow the gap with hyperspectral tracking model. The overall HDSP loss function is the same as the base model without additional adjustments

$$\ell_{\text{track}} = L_{\text{cls}} + \lambda_{\text{iou}} L_{\text{iou}} + \lambda_{L_1} L_1. \tag{19}$$

L_{cls} is the weighted focal loss for classification, L_1, and generalized IoU loss L_{iou} for bounding box regression. λ_{iou} and λ_{L_1} are regularization parameters.

IV. Experiments

A. Experimental Setup

1) HSV Datasets: The datasets are provided by the 2022 hyperspectral object tracking challenge (HOTC) [5]. The entire dataset HOTC consists of 75 HSVs and the corresponding 75 color videos. The training set includes 40 video sequences, while the test set includes 35 video sequences. Each video contains hyperspectral, false color, and RGB videos

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:06:08 UTC from IEEE Xplore. Restrictions apply.

captured in the same scene. RGB videos are obtained with a viewpoint close to the HSVs and at the same frame rate. False-color videos are generated from HSVs using the CIE CMFs. Each hyperspectral image consists of N bands (16 in this dataset) covering the range of 470–620 nm, with each band's size being 512 × 256 pixels. Each video is annotated with challenges related to 11 attributes, including illumination variation (IV), scale variation (SV), occlusion (OCC), deformation (DEF), motion blur (MB), fast motion (FM), in-plane rotation (IPR), out-of-plane rotation (OPR), out-of-view (OV), background clutters (BCs), and low resolution (LR).

2) Evaluation Metrics: The following experiments used four commonly object tracking evaluation metrics, including success plot, precision plot, AUC, and average distance precision score at a 20-pixel threshold (DP@20pixels). The success plot visually showcases the performance of object tracking algorithms at different success thresholds. The precision plot balances the accuracy and recall of the classifier, where a higher AUC indicates superior performance. The DP examines the average distance between the model's detected object position and the actual object position, measured with a 20-pixel threshold to assess distance accuracy. All the results are presented through a one-pass evaluation (OPE), where the tracker runs throughout the entire test sequence and initializes from the bounding box in the initial frame.

3) Implementation Details: The proposed method was implemented in Python using the PyTorch framework and trained with accelerated GPU. The batch size is set to 14, and the AdamW optimizer with a weight decay of 10^{-4} is used. The learning rate is set to 4×10^{-5}. We trained for 50 epochs. The fixed parameters are initialized using a pretrained base model OSTrack [24], while the trainable feature learning parameters are initialized using the Xavier uniform initialization scheme.

B. Comparison to State-of-the-Art Trackers

1) Comparison to State-of-the-Art Hyperspectral Trackers: We compared the proposed HDSP with eight state-of-the-art hyperspectral trackers, including Trans-DAT [42], SEE-Net [39], SST-Net [48], MFI [32], MHT [5], BAE-Net [10], DeepHKCF [35], and CNHT [31]. Among these, MFI, MHT, and CNHT use handcrafted features for tracking, while Trans-DAT, SEE-Net, SST-Net, BAE-Net, and DeepHKCF use deep learning features. As shown in Fig. 5, our HDSP ranks first among all the compared trackers, achieving an accuracy of 0.939 and an AUC score of 0.687. This confirms the effectiveness of HDSP in hyperspectralobject tracking.

2) Comparison to Color Trackers: To better illustrate the effectiveness of the proposed method, we compared HDSP with eight state-of-the-art color trackers, namely, ARTrack [25], SeqTrack [26], OSTrack [24], SiamBAN [18], TransT [22], SiamCAR [49], SiamGAT [17], and STARK [21]. For a fair comparison, the color trackers are compared on false-color videos converted from HSVs via CIE CMFs. Fig. 6 displays the comparison results. Our method achieves the best performance in terms of success and precision.

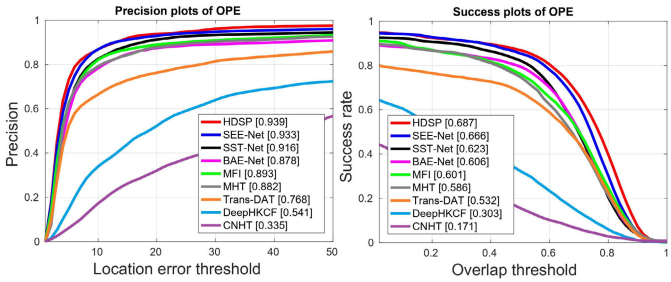


Fig. 5. Precision and success plots' comparison to state-of-the-art hyperspectral trackers on the HOTC dataset.

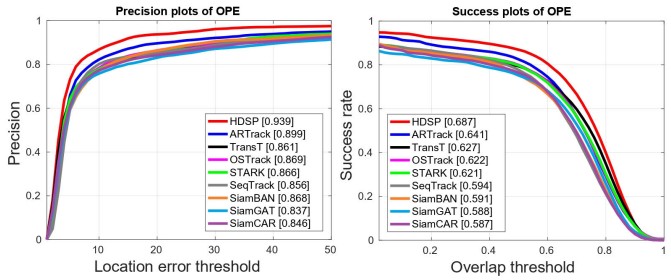


Fig. 6. Precision and success plots' comparison to state-of-the-art color trackers on false-color videos on the HOTC dataset.

3) Attribute-Based Comparison to Hyperspectral Trackers: We compared HDSP with eight hyperspectral trackers on 11 attributes. As shown in Fig. 7, except BC, our HDSP tracker ranks first in ten out of 11 attributes. In the BC scene, due to the possible noise and distortion in the image, the spectral information may cause some interference, which leads to poor performance of HDSP. HDSP performs well in IV, SV, DEF, and OV scenes because the spatiotemporal information provides continuous spatiotemporal features that help adapt to drastic deformation and background changes. In LR, FM, IPR, and MB scenarios, the spectral information helps distinguish the object from the background due to its material discrimination ability. Thus, HDSP provides better tracking in these scenarios as well.

4) Attribute-Based Comparison to Color Trackers: We compared HDSP with eight state-of-the-art color trackers on 11 attributes. Table I displays the comparison results on false-color videos. HDSP achieves the best results except for OV, which further proves the tracking effectiveness of HDSP. ARTrack and OSTrack are more robust to changes in the shape, size, and appearance of objects and are therefore better able to handle objects that are out of view or deformed. Different from other color trackers, HDSP introduces auxiliary modal input of spectral information and spatiotemporal information. Spectral information provides the characteristics of different substances and helps distinguish objects from complex backgrounds. Spatiotemporal information provides the characteristics of object position change, so that the tracking system can quickly and dynamically adapt to the appearance change of the object. Thus, HDSP performs the best under the remaining ten attributes.

5) Visualization Tracking Comparison: To more fully demonstrate the advantages of HDSP, the visual results of seven hyperspectral trackers are shown, including HDSP,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:06:08 UTC from IEEE Xplore. Restrictions apply.

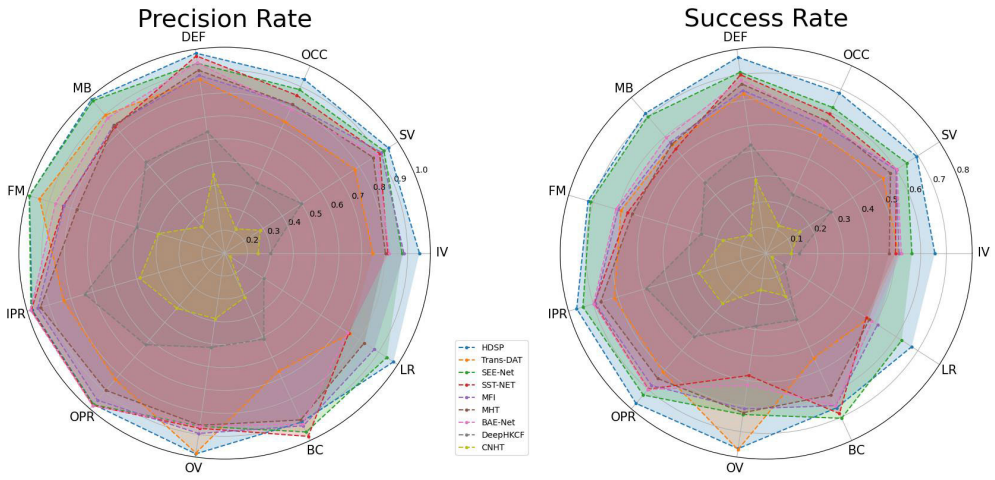


Fig. 7. Attribute-based comparison to state-of-the-art hyperspectral trackers on the HOTC dataset.

TABLE I

ATTRIBUTE-BASED COMPARISON ON HSVS/FALSE-COLOR VIDEOS (EVALUATION METRIC: AUC). THE BEST RESULTS ARE MARKED IN BOLD

Method	IV	SV	OCC	DEF	MB	FM	IPR	OPR	OV	BC	LR	AVG
HDSP (Ours)	**0.655**	**0.695**	**0.684**	**0.769**	**0.718**	**0.721**	**0.767**	**0.772**	0.767	**0.660**	**0.672**	**0.714**
ARTrack	0.574	0.651	0.621	0.721	0.665	0.675	0.751	0.731	**0.841**	0.630	0.600	0.678
SeqTrack	0.554	0.621	0.596	0.693	0.611	0.639	0.664	0.710	0.757	0.566	0.529	0.631
OSTrack	0.596	0.625	0.633	0.744	0.682	0.679	0.700	0.710	0.824	0.574	0.571	0.711
STARK	0.562	0.636	0.600	0.713	0.683	0.645	0.736	0.736	0.769	0.596	0.546	0.692
TransT	0.543	0.634	0.592	0.714	0.645	0.612	0.749	0.745	0.769	0.627	0.523	0.697
SiamCAR	0.542	0.595	0.572	0.696	0.664	0.661	0.651	0.679	0.634	0.567	0.554	0.669
SiamGAT	0.536	0.605	0.581	0.675	0.583	0.598	0.648	0.680	0.619	0.568	0.510	0.636
SiamBAN	0.510	0.608	0.562	0.686	0.673	0.638	0.692	0.707	0.620	0.565	0.573	0.631

Trans-DAT, SEE-Net, SST-Net, MFI, MHT, and BAE-Net. As shown in Fig. 8, the proposed HDSP can better track and deal with challenging scenes. In scenes with fast movement of objects such as *basketball, campus*, and *car3*, due to the lack of consideration of spatiotemporal information, Trans-DAT, SEE-Net, SST-Net, MFI, MHT, and BAE-Net have the problem of losing objects. In challenging scenes such as *coke, forest2*, and *student* with rotation, occlusion, and complex background, because spectral information helps provide detailed information about the composition of substances in the image, it can be applied to distinguish objects from complex backgrounds. Unlike the visual comparison of different frames shown in Fig. 8, to demonstrate the accuracy of object tracking between consecutive frames, we calculated the intersection over union (IoU) between the tracker candidate box and the ground-truth box. As shown in Fig. 9, compared with other trackers, HDSP exhibits higher and more stable IoU values in similar consecutive frames. This further indicates the effectiveness of HDSP in object tracking.

6) Performance Comparison of FPS, Params and FLOPs: We report the FPS of all hyperspectral trackers and the Params and FLOPs for the trackers based on deep learning. As shown in Table II, HDSP achieves the second-fastest tracking speed among competing hyperspectral trackers. The deep learning trackers operated on a machine with an NVIDIA 2080ti GPU. When the number of CPLs is set to 12, the trainable parameter size in HDSP is 253 M. Trans-DAT has the fastest speed and lowest FLOPs, which makes it the most lightweight model; however its performance on the dataset is not as good as our method. SST-Net and BAE-Net use different preprocessing methods to decompose each frame of hyperspectral images into five sets of three-channel images. These sets are individually tracked, with shared model parameters during tracking, and the results from all five sets are aggregated. Consequently, SST-Net and BAE-Net have fewer trainable parameters, slower processing speeds, and higher FLOPs. HDSP and SEE-Net convert each frame of hyperspectral image tracking into a problem of tracking a single three-channel image. Therefore, HDSP and SEE-Net have relatively fewer FLOPs. In addition, some lightweight color trackers [27], [28], [29], [30] have been proposed recently, but these methods differ significantly from hyperspectral trackers in their processing workflows. Specifically, some hyperspectral trackers require preprocessing of hyperspectral images, while others extract features directly in the main network without preprocessing. Therefore, the speed of the existing lightweight color trackers cannot be directly compared with that of hyperspectral trackers.

C. Ablation Study

1) Impact of BSM: We conducted experiments on three different band selection methods: random selection (RS), maximum image entropy selection (MIS), and BSM. It is well-known that in hyperspectral images, the larger the interval between band sequences, the weaker the correlation between bands. This results in a reduction in redundancy among

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:06:08 UTC from IEEE Xplore. Restrictions apply.

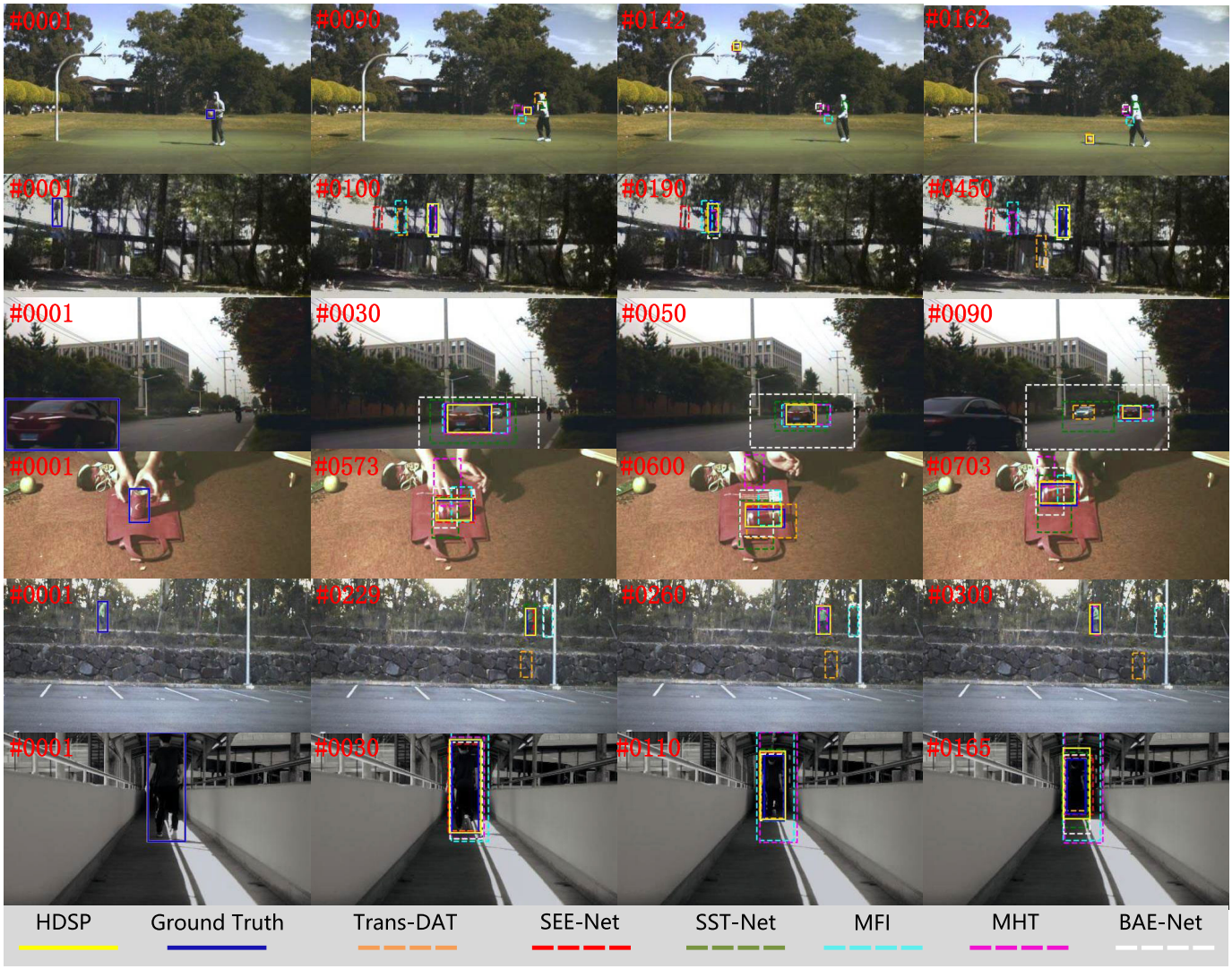


Fig. 8. Visualization of tracking results on six video sequences (i.e., *basketball, campus, car3, coke, forest2*, and *student*). HDSP performs favorably against these trackers. (For better illustration, we show the bounding boxes on the false-color images converted from hyperspectral images.)

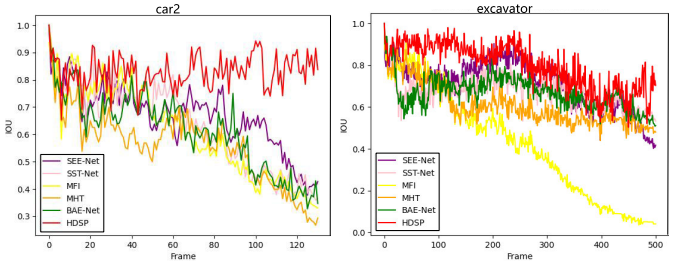


Fig. 9. Comparison of HDSP with state-of-the-art hyperspectral trackers on IoU per frame on two video sequences (i.e., *car2* and *excavator*).

TABLE II

COMPARISON OF HDSP TO STATE-OF-THE-ART TRACKERS ON FPS, PARAMS, AND FLOPS. THE BEST RESULTS ARE MARKED IN BOLD

Device	GPU (FPS)					CPU (FPS)			
Method	HDSP	Trans-DAT	SEE-Net	SST-Net	BAE-Net	MFI	MHT	DeepHKCF	CNHT
FPS	15.60	**27.16**	12.15	0.72	0.87	0.45	2.35	1.02	2.82
Params (M)	253	27.98	206.1	17.8	**17.7**	-	-	-	-
FLOPs (T)	0.23	**0.03**	0.29	1.65	1.48	-	-	-	-

the bands. Therefore, when using RS for band selection, we choose bands with larger intervals in the band sequence, such as the first, fifth, and 15th bands, as the RS results. When using MIS for band selection, we choose the top three bands with the highest entropy as the output. As shown in Table III, BSM can choose bands with the maximum information and minimal redundancy.

2) Number of CPLs: The CPL is used to generate spectral–spatiotemporal prompts stream, which can enhance the characteristics of the foundation information. Therefore, the number of cross-correlated prompt layers is crucial. Table IV outlines the performance of methods using different numbers of CPLs. When there is only ICPL, the number of CPL is one. As the number of layers increases, performance improves, with speed consistently outperforming other hyperspectral trackers. Therefore, in other experiments, we set the layer count to 12.

3) Impact of Dual-Stream Prompt: HDSP is a dual-stream architecture in which the spectral prompt stream supplements detailed information about the composition of objects

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:06:08 UTC from IEEE Xplore. Restrictions apply.

TABLE III

PERFORMANCE OF DIFFERENT BAND SELECTION METHODS. THE BEST RESULTS ARE MARKED IN BOLD

BS Method	RS	MIS	BSM	AUC	DP
Baseline + MDVP	✓			0.657	0.882
Baseline + MDVP		✓		0.666	0.918
Baseline + MDVP			✓	**0.687**	**0.939**

TABLE IV

PERFORMANCE OF OUR METHOD ON THE TEST OF HOTC IS EVALUATED BY VARYING THE NUMBER OF CPLS. THE BEST RESULTS ARE MARKED IN BOLD

Layers	1	2	4	6	8	12
AUC	0.641	0.659	0.668	0.672	0.679	**0.687**
DP	0.885	0.912	0.923	0.930	0.935	**0.939**
FPS	**39.92**	35.15	30.72	26.34	23.70	15.60

TABLE V

PERFORMANCE OF OUR METHOD ON THE TEST OF HOTC WITH VARYING PROMPT ARCHITECTURES. THE BEST RESULTS ARE MARKED IN BOLD

Prompt Stream	Spectral	Spatiotemporal	ALL	AUC	DP
Baseline (OSTrack)				0.627	0.869
Baseline + MCP	✓			0.634	0.886
Baseline + MCP		✓		0.641	0.891
Baseline + MCP			✓	0.660	0.903
Baseline + MDVP			✓	**0.687**	**0.939**

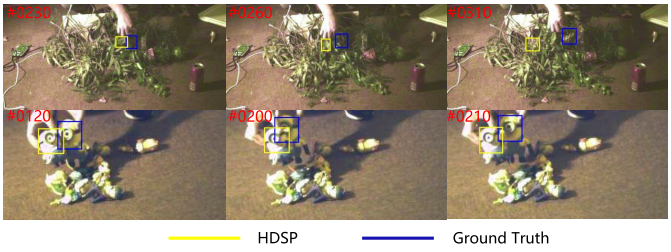


Fig. 10. Failure cases of HDSP (i.e., *fruit* and *toy1*).

in the scene, while the spatiotemporal prompt stream provides object motion information. In Table V, we compare various input streams to assess the effectiveness of spatiotemporal and spectral prompts. Since MDVP is incompatible with a single-stream architecture, we substitute it with the modality-complementary prompter (MCP) from ViPT [13] to ensure fairness when using only the spatiotemporal or spectral prompt stream. In both the cases, the tracking success rate and accuracy surpass the baseline. When using the spatiotemporal prompts stream or the spectral prompts stream as the input stream at the same time, the performance is further improved. Finally, a comparison between MCP and MDVP in dual-stream prompting reveals that MDVP delivers superior performance, indicating its stronger ability to integrate features and generate more robust prompt information. These results affirm the utility of the spatiotemporal prompts and spectral prompts and the effectiveness of the proposed MDVP.

D. Failure Cases

We show cases where HDSP fails to track in sequence *fruit* and *toy1*. As shown in Fig. 10, in sequence *fruit*, it is difficult for our classifier to distinguish the precise location of the object because the foreground is very similar to the background. In sequence *toy1*, the reason for the failure may be that the object is not only occluded but also disturbed by similar objects, which leads to the drift phenomenon in our tracker. These failure cases highlight the challenges of object tracking in complex environments and also provide us with valuable clues to improve the algorithm.

V. CONCLUSION

In this study, we introduce an innovative dual-stream prompt framework to enhance hyperspectral object visual tracking. This framework combines multimodal inputs of spectral and spatiotemporal information to both obtain detailed object of various materials and capture features related to changes in object position, which can facilitate dynamic adaptation to changes in object appearance. In this way, the proposed multimodal spectral–spatiotemporal dual-stream visual prompter learns quickly with a few training parameters, which can improve the accuracy of object classification and regression. The experimental results show that the performance of the proposed method is significantly better than the current state-of-the-art methods and demonstrate the efficiency and applicability of the proposed prompting framework.

APPENDIX A
CIE COLOR MATCHING FUNCTIONS

CIE color matching functions (CMFs) are a set of standard functions used to describe the perceptual weights of the human eye to different wavelengths of light. It can be used to convert hyperspectral images into CIE *XYZ* color space, and then generate false-color images. The CIE CMFs usage mentioned in this article comes from MHT [5]. It is mainly divided into the following steps.

1) Given an hyperspectral image $\mathbf{H}_t \in \mathbb{R}^{N\times H\times W}$ with $H \times W$ pixels and N bands, this step converts the hyperspectral image to a CIE *XYZ* image

$$\mathbf{I} = \sum_{i=1}^{N} \mathbf{H}_t(:, i)\mathbf{A}(i, :) \tag{20}$$

where $\mathbf{A} \in \mathbb{R}^{N\times 3}$ are the CMFs. Here we used 10-deg *XYZ* CMFs transformed from the CIE (2006) with a step size of 0.1 nm. After that, $\mathbf{I}$ is converted into the default RGB color space sRGB.

2) The color transformation method in [50] is used to make the intensity of the converted false-color image $\mathbf{F}_t \in \mathbb{R}^{3\times H\times W}$ close to the color intensity of the RGB image.

APPENDIX B
FALSE-COLOR IMAGE AND PSEUDOCOLOR IMAGE

False-color images and pseudocolor images are images generated by different processing methods of hyperspectral images. Specifically, pseudocolor images usually map the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:06:08 UTC from IEEE Xplore. Restrictions apply.

bands of the invisible spectrum to the red, green, and blue channels of the visible spectrum for easy observation and analysis by the human eye. The pseudocolor image mentioned in this article is a three-channel color image synthesized by different bands. Through the standard color conversion model, the false-color image converts the hyperspectral data into a visually close to the real color image, so that it is more in line with the color perception of the human eye. The false-color image mentioned in this article is a three-channel image converted from a hyperspectral image using the `CMFs`.

REFERENCES

[1] K. A. Joshi and D. G. Thakore, "A survey on moving object detection and tracking in video surveillance system," *Int. J. Soft Comput. Eng.*, vol. 2, no. 3, pp. 44–48, 2012.

[2] N. Liu, P. Shapira, and X. Yue, "Tracking developments in artificial intelligence research: Constructing and applying a new search strategy," *Scientometrics*, vol. 126, no. 4, pp. 3153–3192, Apr. 2021.

[3] G. Barquero, I. Hupont, and C. F. Tena, "Rank-based verification for long-term face tracking in crowded scenes," *IEEE Trans. Biometrics, Behav., Identity Sci.*, vol. 3, no. 4, pp. 495–505, Oct. 2021.

[4] B. Li, W. Wu, Q. Wang, F. Zhang, J. Xing, and J. Yan, "SiamRPN++: Evolution of Siamese visual tracking with very deep networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 4282–4291.

[5] F. Xiong, J. Zhou, and Y. Qian, "Material based object tracking in hyperspectral videos," *IEEE Trans. Image Process.*, vol. 29, pp. 3719–3733, 2020.

[6] K. Qian, J. Shen, S. Wang, and W. Sun, "Recent advances in object tracking using hyperspectral videos: A survey," *Multimedia Tools Appl.*, vol. 83, no. 18, pp. 56155–56181, Dec. 2023.

[7] Y. Tang, Y. Liu, L. Ji, and H. Huang, "Robust hyperspectral object tracking by exploiting background-aware spectral information with band selection network," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[8] Y. Tang, H. Huang, Y. Liu, and Y. Li, "A Siamese network-based tracking framework for hyperspectral video," *Neural Comput. Appl.*, vol. 35, no. 3, pp. 2381–2397, Jan. 2023.

[9] Z. Liu, X. Wang, Y. Zhong, M. Shu, and C. Sun, "SiamHYPER: Learning a hyperspectral object tracker from an RGB-based tracker," *IEEE Trans. Image Process.*, vol. 31, pp. 7116–7129, 2022.

[10] Z. Li, F. Xiong, J. Zhou, J. Wang, J. Lu, and Y. Qian, "BAE-Net: A band attention aware ensemble network for hyperspectral object tracking," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2020, pp. 2106–2110.

[11] L. Gao et al., "CBFF-Net: A new framework for efficient and accurate hyperspectral object tracking," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5506114.

[12] W. Li, Z. Hou, J. Zhou, and R. Tao, "SiamBAG: Band attention grouping-based Siamese object tracking network for hyperspectral videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514712.

[13] J. Zhu, S. Lai, X. Chen, D. Wang, and H. Lu, "Visual prompt multi-modal tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 9516–9526.

[14] J. Yang, Z. Li, F. Zheng, A. Leonardis, and J. Song, "Prompting for multi-modal tracking," in *Proc. 30th ACM Int. Conf. Multimedia*, Oct. 2022, pp. 3492–3500.

[15] A. Lukezic, T. Vojir, L. C. Zajc, J. Matas, and M. Kristan, "Discriminative correlation filter with channel and spatial reliability," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6309–6318.

[16] J. F. Henriques, R. Caseiro, P. Martins, and J. Batista, "High-speed tracking with kernelized correlation filters," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 3, pp. 583–596, Mar. 2015.

[17] D. Guo, Y. Shao, Y. Cui, Z. Wang, L. Zhang, and C. Shen, "Graph attention tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 9543–9552.

[18] Z. Chen et al., "SiamBAN: Target-aware tracking with Siamese box adaptive network," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 4, pp. 5158–5173, Apr. 2022.

[19] A. Vaswani et al., "Attention is all you need," 2017, *arXiv:1706.03762*.

[20] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 213–229.

[21] B. Yan, H. Peng, J. Fu, D. Wang, and H. Lu, "Learning spatio-temporal transformer for visual tracking," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10448–10457.

[22] X. Chen, B. Yan, J. Zhu, D. Wang, X. Yang, and H. Lu, "Transformer tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 8126–8135.

[23] Z. Fu, Z. Fu, Q. Liu, W. Cai, and Y. Wang, "SparseTT: Visual tracking with sparse transformers," 2022, *arXiv:2205.03776*.

[24] B. Ye, H. Chang, B. Ma, S. Shan, and X. Chen, "Joint feature learning and relation modeling for tracking: A one-stream framework," in *Proc. 17th Eur. Conf. Comput. Vis. (ECCV)*. Cham, Switzerland: Springer, Oct. 2022, pp. 341–357.

[25] X. Wei, Y. Bai, Y. Zheng, D. Shi, and Y. Gong, "Autoregressive visual tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 9697–9706.

[26] X. Chen, H. Peng, D. Wang, H. Lu, and H. Hu, "SeqTrack: Sequence to sequence learning for visual object tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2023, pp. 14572–14581.

[27] V. Borsuk, R. Vei, O. Kupyn, T. Martyniuk, I. Krashenyi, and J. Matas, "FEAR: Fast, efficient, accurate and robust visual tracker," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2022, pp. 644–663.

[28] X. Chen, B. Kang, D. Wang, D. Li, and H. Lu, "Efficient visual tracking via hierarchical cross-attention transformer," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2022, pp. 461–477.

[29] P. Blatter, M. Kanakis, M. Danelljan, and L. V. Gool, "Efficient visual tracking with exemplar transformers," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2023, pp. 1571–1581.

[30] B. Kang, X. Chen, D. Wang, H. Peng, and H. Lu, "Exploring lightweight hierarchical vision transformers for efficient visual tracking," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 9612–9621.

[31] K. Qian, J. Zhou, F. Xiong, H. Zhou, and J. Du, "Object tracking in hyperspectral videos with convolutional features and kernelized correlation filter," in *Proc. 1st Int. Conf. Smart Multimedia (ICSM)*, Toulon, France. Cham, Switzerland: Springer, Aug. 2018, pp. 308–319.

[32] Z. Zhang, K. Qian, J. Du, and H. Zhou, "Multi-features integration based hyperspectral videos tracker," in *Proc. 11th Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens. (WHISPERS)*, Mar. 2021, pp. 1–5.

[33] Y. Zhang, X. Li, B. Wei, L. Li, and S. Yue, "A fast hyperspectral tracking method via channel selection," *Remote Sens.*, vol. 15, no. 6, p. 1557, Mar. 2023.

[34] H. K. Galoogahi, A. Fagg, and S. Lucey, "Learning background-aware correlation filters for visual tracking," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 1135–1143.

[35] B. Uzkent, A. Rangnekar, and M. J. Hoffman, "Tracking in aerial hyperspectral videos using deep kernelized correlation filters," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 449–461, Jan. 2019.

[36] S. Wang, K. Qian, and P. Chen, "BS-SiamRPN: Hyperspectral video tracking based on band selection and the Siamese region proposal network," in *Proc. 12th Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens. (WHISPERS)*, Sep. 2022, pp. 1–8.

[37] C. Zhao, H. Liu, N. Su, and Y. Yan, "TFTN: A transformer-based fusion tracking framework of hyperspectral and RGB," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5542515.

[38] L. Gao, L. Chen, P. Liu, Y. Jiang, W. Xie, and Y. Li, "A transformer-based network for hyperspectral object tracking," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5528211.

[39] Z. Li, F. Xiong, J. Zhou, J. Lu, and Y. Qian, "Learning a deep ensemble network with band importance for hyperspectral object tracking," *IEEE Trans. Image Process.*, vol. 32, pp. 2901–2914, 2023.

[40] Y. Su, S. Mei, G. Zhang, Y. Wang, M. He, and Q. Du, "Gaussian information entropy based band reduction for unsupervised hyperspectral video tracking," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2022, pp. 791–794.

[41] X. Jiang, X. Wang, C. Sun, Z. Zhu, and Y. Zhong, "A channel adaptive dual Siamese network for hyperspectral object tracking," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5403912.

[42] Y. Wu, L. Jiao, X. Liu, F. Liu, S. Yang, and L. Li, "Domain adaptation-aware transformer for hyperspectral object tracking," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 9, pp. 8041–8052, Sep. 2024.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:06:08 UTC from IEEE Xplore. Restrictions apply.

[43] M. Jia et al., "Visual prompt tuning," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, Oct. 2022, pp. 709–727.
[44] T. Yang, Y. Zhu, Y. Xie, A. Zhang, C. Chen, and M. Li, "AIM: Adapting image models for efficient video action recognition," 2023, *arXiv:2302.03024*.
[45] J. Wang et al., "Multi-modal visual tracking based on textual generation," *Inf. Fusion*, vol. 112, Dec. 2024, Art. no. 102531.
[46] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.
[47] J. Sander, M. Ester, H.-P. Kriegel, and X. Xu, "Density-based clustering in spatial databases: The algorithm GDBSCAN and its applications," *Data Min. Knowl. Discov.*, vol. 2, no. 2, pp. 169–194, 1998.
[48] Z. Li, X. Ye, F. Xiong, J. Lu, J. Zhou, and Y. Qian, "Spectral–spatial–temporal attention network for hyperspectral tracking," in *Proc. 11th Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens. (WHISPERS)*, Mar. 2021, pp. 1–5.
[49] D. Guo, J. Wang, Y. Cui, Z. Wang, and S. Chen, "SiamCAR: Siamese fully convolutional classification and regression for visual tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 6269–6277.
[50] F. Pitie and A. Kokaram, "The linear Monge-Kantorovitch linear colour mapping for example-based colour transfer," in *Proc. 4th Eur. Conf. Vis. Media Prod.*, London, U.K., 2007, pp. 1–9, doi: 10.1049/cp:20070055.



Rui Yao received the Ph.D. degree in computer science from the Northwestern Polytechnical University, Xi'an, China, in 2013.

From 2011 to 2012 and 2016 to 2017, he conducted academic research with the University of Adelaide, SA, Australia. He is currently a Professor with the School of Computer Science and Technology, China University of Mining and Technology, Xuzhou, China. His research interests include computer vision and machine learning.



Lu Zhang received the B.S. degree from Wuxi University, Wuxi, China, in 2022. She is currently pursuing the master's degree with the School of Computer Science and Technology, China University of Mining and Technology, Xuzhou, China.

Her research interests include computer vision, pattern recognition, and hyperspectral object tracking.



Yong Zhou received the M.S. and Ph.D. degrees in control theory and control engineering from China University of Mining and Technology, Xuzhou, China, in 2003 and 2006, respectively.

He is currently a Professor with the School of Computer Science and Technology, China University of Mining and Technology. His research interests include machine learning, intelligence optimization, and data mining.



Hancheng Zhu received the B.S. degree from Changzhou Institute of Technology, Changzhou, China, in 2012, and the M.S. and Ph.D. degrees from China University of Mining and Technology, Xuzhou, China, in 2015 and 2020, respectively.

He is currently a tenure-track Associate Professor with the School of Computer Science and Technology, China University of Mining and Technology. His research interests include image esthetics assessment and affective computing.



Jiaqi Zhao (Member, IEEE) received the B.S. and the Ph.D. degrees from Xidian University, Xi'an, China, in 2010 and 2017, respectively.

From 2013 to 2014, he was an Exchange Ph.D. Student with Leiden Institute for Advanced Computer Science (LIACS), Leiden University, Leiden, The Netherlands. He is currently an Associate Professor with China University of Mining and Technology, Xuzhou, China. His research interests include multiobjective learning and computer vision.



Zhiwen Shao received the B.Eng. degree in computer science and technology from the Northwestern Polytechnical University, Xi'an, China, in 2015, and the Ph.D. degree from Shanghai Jiao Tong University, Shanghai, China, in 2020.

He is currently an Associate Professor with the School of Computer Science and Technology, China University of Mining and Technology, Xuzhou, China. From 2017 to 2018, he was a joint Ph.D. Student with the Multimedia and Interactive Computing Laboratory, Nanyang Technological University, Singapore. His research interests include computer vision and deep learning.

Dr. Shao has been serving as a PC member in IJCAI and AAAI.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:06:08 UTC from IEEE Xplore. Restrictions apply.